Seasonal Variation in Spatial Extremal Dependence of Wind Gusts in the Netherlands

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Introduction

The Netherlands is situated within a low-lying river delta, with a substantial percentage of its land located below sea level. The country's relationship with extreme weather has historically been framed as a relationship with water: the flood of February 1953, known in Dutch as the *Watersnoodramp*, killed more than 1800 people in Zeeland and South Holland after dikes failed during a spring tide combined with a severe North Sea storm (Gerritsen, 2005). The subsequent Delta Plan and Dutch flood-protection regime implemented safety standards, with historical return periods ranging from 1250 years for major river areas to 10 000 years for the most densely populated coastal parts of Holland (Deltacommissie, 1960). These return periods illustrate how extreme-value reasoning has long been embedded in Dutch climate-risk management. Wind has received comparatively less attention in this Dutch climate-risk narrative, despite the fact that 10 out of 20 most costly climate related hazards were windstorms (Verbond van Verzekeraars, 2017). Investigating seasonal differences in spatial extremal dependence is relevant because the societal impact of wind extremes depends not only on the magnitude of gusts at individual locations, but also on the spatial extent of the event.

Existing windstorm studies often focus on marginal extremes, return levels, or losses. For national risk, however, the relevant question is not only how large the gust is at one station, but how many other stations are extreme at the same time. This thesis addresses this gap by focusing on pairwise extremal dependence rather than only marginal wind-gust behaviour.

Wind extremes are spatial events. A severe winter storm associated with a mature extratropical cyclone may produce damaging gusts across large parts of the Netherlands within a few hours, reflecting the large spatial footprints commonly associated with European windstorms (Sharkey et al., 2020; Little et al., 2023). By contrast, summer convective gusts are often linked to thunderstorms and local-scale downbursts, whose small-scale and non-stationary nature can generate sharp spatial differences in observed gust intensity (Mohr et al., 2017). These two events may look similar from the perspective of a single-station maximum, but they imply very different risks for society. The first is a system-wide event, with simultaneous pressure on infrastructure, insurers, transport networks and emergency services; the second is more localised, but potentially harder to anticipate at fine spatial scales (Gliksman et al., 2023; Fonseca Cerda et al., 2025). This distinction motivates the focus on spatial extremal dependence rather than only marginal wind-gust extremes.

Spatial extreme value theory provides a general framework for analysing such tail dependencies. The pairwise extremal coefficient $\theta_{ij}$and the upper tail dependence coefficient χ_{ij} summarise the joint behaviour of a station pair at the upper tail (Davison et al., 2012; Cooley, Cisewski, et al., 2012). The extremal coefficient θ_{ij} lies between one and two, with values close to one corresponding to strong extremal dependence and values close to two corresponding to weak extremal dependence. The upper tail dependence coefficient χ_{ij} describes the limiting probability that one station is extreme conditional on another station being extreme. Both coefficients are functions of the distance between stations, and the spatial extremal dependence structure can be summarised as a dependence–distance curve. The methodological problem is therefore not only to describe wind-gust extremes, but to determine whether seasonal differences in spatial tail dependence can be estimated reliably from a finite and spatially irregular station network. Splitting the sample by season reduces the effective number of extreme observations, while estimating dependence for many station pairs introduces additional sampling variability across distances. Therefore, we frame the empirical comparison between winter and summer as a problem of finite-sample inference for spatial extremal dependence. The simulation study is used to examine whether the selected pairwise estimators can recover known differences in extremal dependence under controlled data-generating processes before the same estimators are applied to the KNMI observations.

Research question

The main research question of this thesis is:

Is there more spatial extremal dependence of wind gusts between KNMI station locations in winter than in summer in the Netherlands over the period 1991–2025?

The comparison is between winter, defined as December–February (DJF), and summer, defined as June–August (JJA). The hypothesis is that winter wind-gust extremes are more spatially coherent than summer wind-gust extremes. In terms of the extremal coefficient, this corresponds to lower values of θ_{ij} for winter than for summer at comparable inter-station distances. In terms of the upper-tail dependence coefficient, it corresponds to higher values of χ_{ij}. The empirical analysis therefore compares season-specific dependence–distance curves rather than only testing for a difference at a single station pair.

The empirical application uses hourly observations from the network of automatic weather stations operated by the Royal Netherlands Meteorological Institute (KNMI). The hourly maximum wind gust, denoted FX, is expressed in metres per second. Stations are indexed by their location $s_i \in \mathbb{R}^2$, with latitude, longitude and a sensor-height attribute recorded in the station metadata. The full dataset covers the years from 1951 to 2026. Due to station-network expansion and missing observations, the number of stations reporting FX in any given hour varies with time. The analysis sample is 1991–2025, which is the first 35-year period with a sufficiently large and stable station network.

The contribution of this thesis is to study the finite-sample performance of pairwise estimators of spatial extremal dependence in a seasonal comparison design. Empirically, the thesis applies this framework to a dataset of hourly wind-gust observations from the Netherlands and compares the upper-tail dependence, the extremal coefficient θ and the tail-dependence coefficient χ, across stations in order to determine whether there are significant seasonal differences in spatial extremal dependence.

Literature Review

This section reviews the literature on extreme-value theory for environmental data, spatial extremes and max-stable processes, pairwise extremal-dependence measures, data-generating processes for the simulation study, inference and uncertainty for dependence estimators, and seasonal and non-stationary extremal dependence.

Extreme-value foundations for environmental data

Extreme-value theory provides the statistical framework for studying rare events in the tails of a distribution. For the canonical case of maxima, let $X_1, \dots, X_n$ be independent observations with common distribution function F and let $M_n = \max\{X_1, \dots, X_n\}$. Then

$$\Pr\{M_n \leq z\} = \Pr\{X_1 \leq z, \dots, X_n \leq z\} = F(z)^n.$$

The results of Fisher and Tippett (1928) and Gnedenko (1943) show that, under normalisation, any non-degenerate limit of normalised sample maxima must belong to the generalised extreme value (GEV) family. The Gumbel, Fréchet, and Weibull types are special cases of the GEV distribution, whose shape parameter governs tail behaviour. This result justifies modelling tail observations directly rather than average behaviour, and it provides the theoretical basis for the block-maxima approach, in which the data are divided into non-overlapping blocks and the maximum within each block is modelled. The block-maxima method is also the entry point for spatial extremes, because spatial max-stable processes arise as the limiting objects for spatial block maxima (Laurens de Haan and Pereira, 2006).

In our paper the blocks are days: the hourly wind-gust observations are aggregated to daily maximum wind gusts at each station, and these daily maximum wind gusts are the data object on which all subsequent dependence estimation is performed. Daily maximum wind gusts are a deliberate compromise on the block length. On the one hand, raw hourly gusts are strongly serially dependent, since consecutive hours typically belong to the same passing storm system; aggregating to a daily maximum removes the most immediate within-day temporal clustering and yields blocks that are much closer to the independent-maxima setting that the GEV theory describes. On the other hand, taking the block to be a whole season would leave only one observation per station per season, that is roughly 35 seasonal maxima over 1991–2025, which is far too few to estimate pairwise spatial dependence stably for each of the many station pairs. Daily maximum wind gusts retain on the order of ninety observations per station per season, a sample large enough for the pairwise estimators below while still respecting the block-maxima logic.

A second method is the peaks-over-threshold (POT) framework, which studies observations exceeding a high threshold rather than block maxima. Its theoretical justification is the Pickands–Balkema–de Haan theorem: exceedances above a sufficiently high threshold can be approximated by a generalised Pareto distribution (Balkema and L. de Haan, 1974; Pickands, 1975). POT uses more tail information than block maxima (Brabson and Palutikof, 2000), but its threshold is finite in observed data while the theory concerns an asymptotic approximation, so the threshold choice is itself an inferential decision (S. Coles, 2001). A further difficulty for hourly data is that consecutive exceedances usually belong to the same storm and cannot be treated as independent tail events; correcting for this requires declustering, for instance the automatic interexceedance-time scheme of Ferro and Segers (2003), or explicit temporal modelling of the exceedances, as in the Markov-chain treatment of hourly wind speeds by Fawcett and Walshaw (2006). Because daily aggregation already addresses the most immediate form of this serial dependence, our paper treats POT and declustering as background and as a possible robustness alternative rather than as the main empirical strategy.

The marginal extreme-value step is only the first layer of the research cake. The second layer is spatial: conditional on one station observing an extreme daily maximum wind gust, how likely is it that another station also observes an extreme value on the same day? This shifts the focus from univariate extremes to spatial extremal dependence, whose estimands are pairwise tail-dependence measures, extremal coefficients, and dependence–distance curves, which we review in the next sections.

Spatial extremes and max-stable processes

For block maxima observed over space, max-stable processes provide the standard limiting framework. A max-stable process arises as the limit of pointwise maxima of independent replications of a spatial random field, similar to how the GEV distribution arises as the limit of scalar maxima. Within this framework extremal dependence becomes a spatial object: the joint upper tail is described as a function of location and distance. Laurens de Haan and Pereira (2006) give a foundational treatment of stationary max-stable models for spatial extremes, showing how spatial extremal dependence can be governed by a small number of parameters and related to the distance between locations. The limitation is that stationary max-stable models impose a homogeneous dependence structure, which may be restrictive for Dutch gusts, since winter storms and summer storms do not necessarily share the same spatial dependence pattern.

Davison et al. (2012) review the main statistical approaches to spatial extremes, including latent-variable models, copula models, and max-stable processes. The conclusion from their comparison is that good marginal modelling is not sufficient for modelling spatial extremes: models that fit univariate extremes well may still misrepresent joint extremes across locations. This motivates our focus on extremal-dependence summaries rather than on marginal gust behaviour alone. Likelihood-based inference for max-stable processes is hard, because the full multivariate density is either unavailable or computationally infeasible once the number of locations is moderate. Padoan et al. (2010) replace the full likelihood with a pairwise composite likelihood, making parametric max-stable inference feasible for spatial extremes and establishing pairwise inference as a standard methodological mitigation.

In our paper, max-stable processes serve mainly as a theoretical reference point and as the benchmark data-generating process for the simulation study (Section 2.4), rather than as a model that is fully fitted to the Dutch wind-gust data. We adopt the pairwise logic of Padoan et al. (2010), but the empirical objects are nonparametric pairwise summaries computed directly from the seasonal daily maximum wind gusts, not the parameters of a fitted max-stable family. This keeps the winter–summer comparison transparent and avoids committing the empirical analysis to a single parametric dependence family across inter-station distances.

Pairwise extremal-dependence measures

The empirical analysis requires dependence measures that are interpretable at the station-pair level. Cooley, Cisewski, et al. (2012) survey several tools for measuring spatial extremal dependence, including extremal coefficients, madograms, and tail-dependence summaries. These reduce the high-dimensional spatial dependence problem to interpretable pairwise quantities that can be compared across seasons and plotted against inter-station distance. In our paper all three are estimated from the seasonal daily maximum wind gusts, separately for winter (DJF) and summer (JJA), and the margins are transformed separately by station and season before estimation. Standardising margins station-by-station and season-by-season is important: coastal and inland stations have different local gust climates, and winter and summer differ in their marginal gust distributions, so transforming each station–season margin to a common empirical uniform scale ensures that the winter–summer comparison reflects differences in spatial dependence rather than differences in the marginal distributions.

The extremal coefficient is one of the most common summaries. For two locations s_i and s_j, the pairwise extremal coefficient $\theta(s_i, s_j)$ lies in $[1, 2]$: values close to one correspond to strong extremal dependence, and values close to two to near-independence. In a seasonal comparison, stronger winter dependence corresponds to lower winter values of $\theta(s_i, s_j)$ than summer values at comparable distances.

An empirical route to the extremal coefficient is the F-madogram. Cooley, Naveau, et al. (2006) study madogram-based dependence summaries for spatial max-stable random fields. For transformed margins, the F-madogram between two locations is

$$\nu_F(s_i, s_j) = \tfrac{1}{2}\,\mathbb{E}\left[\left|F_{s_i}\{X(s_i)\} - F_{s_j}\{X(s_j)\}\right|\right],$$

where F_{s_i} and F_{s_j} are the marginal distribution functions at the two locations. Under max-stability, the F-madogram and the pairwise extremal coefficient are linked through

$$\theta(s_i, s_j) = \frac{1 + 2\,\nu_F(s_i, s_j)}{1 - 2\,\nu_F(s_i, s_j)}.$$

The F-madogram gives a direct nonparametric estimate of a dependence measure that can be summarised as a dependence–distance curve, without fitting a parametric max-stable model. The upper-tail dependence coefficient χ provides an additional interpretation. Let $U(s_i)$ and $U(s_j)$ denote marginally standardised observations on a uniform scale. The asymptotic coefficient is

$$\chi(s_i, s_j) = \lim_{u \uparrow 1} \Pr\{U(s_j) > u \mid U(s_i) > u\},$$

provided the limit exists, and the corresponding finite-level quantity is

$$\chi_u(s_i, s_j) = \Pr\{U(s_j) > u \mid U(s_i) > u\}.$$

Here the standardised observations are the seasonal daily maximum wind gusts on the uniform scale, so χ_u is computed within each season from daily co-exceedances. The finite-level χ_u can be read directly off the research question: if one KNMI station records an extreme daily maximum wind gust, how likely is another station to record one on the same day? Larger values of χ_u indicate stronger finite-level co-exceedance, just as smaller values of θ indicate stronger extremal dependence. The coefficients χ and $\bar{\chi}$ were introduced to distinguish asymptotic dependence from asymptotic independence by S. Coles et al. (1999), who also discuss their estimation at finite levels. A caveat is that dependence at observed high levels need not coincide with the limiting asymptotic dependence class. Wadsworth and Tawn (2012) show that max-stable models can be too restrictive when data exhibit dependence at observed levels but become independent in the limit, and Huser and Wadsworth (2022) make the same point about classical max-stable models: they are theoretically natural for asymptotic dependence, but finite environmental data may display weakening dependence at increasingly high levels. This matters for gusts, since winter storms may be spatially extensive while some summer events are localised. Our paper therefore reports finite-level dependence on a small grid of thresholds rather than selecting a single threshold.

Data-generating processes for the simulation study

To validate whether the estimators can recover a true dependence structure, recovery must be measured against a known dependence–distance relation. The pairwise extremal coefficients of max-stable processes are available in closed form, which makes these processes suitable as benchmark data-generating processes: a simulated field carries a known $\theta(s_i, s_j)$ at every distance, against which the estimated curve can be compared. The composite-likelihood machinery of Padoan et al. (2010) is built on exactly these parametric families, and the same families serve here as the known dependence structure to be recovered rather than as an estimation target. The Schlather model (Schlather, 2002) provides a flexible stationary max-stable family with a tractable bivariate dependence function, while the Brown–Resnick process (Brown and Resnick, 1977), characterised through a negative-definite variogram, is a widely used family whose extremal dependence is controlled by that variogram and ranges from strong to near-independent as the variogram range is varied. Dombry et al. (2016) provide exact simulation algorithms for max-stable processes, which avoid the approximation error of naive truncated constructions and ensure that the known dependence structure is the one actually simulated.

Our paper uses the Brown–Resnick process as the main benchmark data-generating process, because its variogram gives direct and interpretable control over how extremal dependence decays with distance, which is exactly the feature the estimators are meant to recover. The null and alternative designs are implemented by choosing one such family and setting its variogram-range parameters so that the two regimes either share a dependence structure (null) or differ by a controlled amount (alternative), with the alternative calibrated so that one regime is more strongly dependent over distance than the other. We stress that the Brown–Resnick process is used only as a controlled benchmark whose true dependence is known; we do not assume that the Dutch wind-gust data are exactly Brown–Resnick. The purpose of the simulation is solely to test whether the pairwise estimators can recover a known winter-versus-summer difference in spatial extremal dependence under a data-generating process where that difference is fixed by design.

Inference and uncertainty for dependence estimators

Because our paper compares estimated dependence–distance curves, the sampling behaviour of the estimators is itself a concern: a difference between two curves is only informative once the uncertainty in each is quantified. Two sources of variability matter. First, the F-madogram and χ_u estimators are empirical means and conditional frequencies computed from a finite number of seasonal daily maximum wind gusts, so their variance grows as the season-split sample shrinks. Second, daily maximum wind gusts are not fully independent in time: although daily aggregation removes the within-day clustering present in hourly data, a single multi-day storm can produce extreme daily maxima on several consecutive days, so consecutive daily observations may still be dependent. Uncertainty quantification should therefore not treat all daily observations as independent.

For this reason our paper quantifies uncertainty by resampling temporal blocks rather than individual days. A year or seasonal-block bootstrap resamples whole years, or whole seasonal segments, with replacement and recomputes the full dependence–distance curves on each resample, so that the resulting variability reflects both the finite sample size and the residual serial dependence between consecutive daily maximum wind gusts within a storm. This block-resampling logic is the daily-maxima counterpart of the cluster-based resampling used for hourly exceedances: the interexceedance-time and declustering machinery of Ferro and Segers (2003), and the temporal modelling of Fawcett and Walshaw (2006), address the same serial-dependence problem at the hourly-exceedance level and remain available as robustness checks. For the spatial estimators, the pairwise composite-likelihood framework of Padoan et al. (2010) comes with asymptotic standard errors, but these rely on the parametric model being correctly specified; the nonparametric pairwise summaries used here are instead assessed by block resampling and, in the simulation, by direct Monte Carlo evaluation against the known truth. The simulation study thus plays the role that an analytic sampling distribution would play if one were available, characterising bias and variability as a function of sample size, distance, and threshold.

Seasonal and non-stationary extremal dependence

The third layer of the research cake can be viewed as a restricted form of non-stationary spatial extremal dependence: instead of one dependence structure for all daily maximum wind gusts, our paper allows the dependence relation to differ between two regimes, with season treated as a simple binary covariate for the dependence structure. Huser and Genton (2016) develop non-stationary dependence structures for spatial extremes, showing that extremal dependence itself, not only the marginal distribution, can vary with covariates. The seasonal split is a deliberately simple version of this idea, and it also addresses a stationarity concern. Many extreme-value methods rely on stationarity, whereas pooled daily maximum wind gusts combine different meteorological regimes; estimating one dependence curve from the unstratified daily maxima would mix large-scale winter storms with more localised summer convective events. Stratifying by season does not remove all non-stationarity, but it makes the working assumption more credible and gives each estimated dependence–distance curve a clearer interpretation.

The literature above serves distinct methodological roles. The classical results of Fisher and Tippett (1928), Gnedenko (1943), Balkema and L. de Haan (1974), and Pickands (1975) justify modelling tail observations and motivate the daily block-maxima choice, while Ferro and Segers (2003) and Fawcett and Walshaw (2006) supply the temporal-dependence corrections that would be needed if the analysis worked from hourly exceedances rather than daily maximum wind gusts. The spatial-extremes work of Laurens de Haan and Pereira (2006), Davison et al. (2012), and Padoan et al. (2010) provides the max-stable and pairwise-inference framework, and Cooley, Naveau, et al. (2006)

and Cooley, Cisewski, et al. (2012) motivate the pairwise estimands – the extremal coefficient, the F-madogram, and dependence–distance curves – estimated here from seasonal daily maximum wind gusts. S. Coles et al. (1999) underpins the finite-level χ_u diagnostic, while Wadsworth and Tawn (2012) and Huser and Wadsworth (2022) caution that finite-level dependence may differ from asymptotic dependence, motivating the threshold-grid robustness check. The benchmark data-generating processes for the validation study draw on the max-stable families of Padoan et al. (2010) and, in particular, the Brown–Resnick process (Brown and Resnick, 1977) with exact simulation (Dombry et al., 2016).

The applied studies of Bonazzi et al. (2012), Della-Marta et al. (2009), Roberts et al. (2014), Klawa and Ulbrich (2003), and S. G. Coles and Walshaw (1994) supply the context: windstorms are spatial events, gust fields require local standardisation, and wind extremes may depend on storm structure and direction. Huser and Genton (2016) shows that extremal dependence can itself be non-stationary. The contribution of our paper is to combine these ideas in a deliberately narrow, methodologically focused setting: to establish, by simulation, when a seasonal difference in spatial extremal dependence can be recovered from finite daily maximum wind gusts on a fixed, irregular station network, and then to apply the validated estimators to Dutch wind gusts. Our paper does not build a compound-hazard model, fit a high-dimensional vine copula, or attribute changes to climate change; its object is the finite-sample comparison of winter and summer dependence–distance curves and finite-level tail-dependence probabilities across KNMI station pairs.

Mathematical details for pairwise extremal dependence estimators

Probability integral transform

We prove that the marginal transform $U_{i,t} = F_i(X_{i,t})$ has a standard uniform distribution when F_i is continuous.

Proof. Let $X_{i,t}$ be a real-valued random variable with continuous distribution function F_i. Define

$$U_{i,t} = F_i(X_{i,t}).$$

For $u < 0$,

$$\Pr(U_{i,t} \le u) = 0.$$

For $u > 1$,

$$\Pr(U_{i,t} \le u) = 1.$$

Let $u \in [0,1]$. Define the generalized inverse

$$F_i^{-1}(u) = \inf\{x \in \mathbb{R} : F_i(x) \ge u\}.$$

Since F_i is continuous,

$$F_i(F_i^{-1}(u)) = u.$$

Then

$$\begin{aligned}
\Pr(U_{i,t} \le u) &= \Pr(F_i(X_{i,t}) \le u) \\
&= \Pr(X_{i,t} \le F_i^{-1}(u)) \\
&= F_i(F_i^{-1}(u)) \\
&= u.
\end{aligned}$$

Hence

$$\Pr(U_{i,t} \le u) = u, \qquad u \in [0,1].$$

Therefore

$$U_{i,t} \sim \mathrm{Uniform}(0,1).$$

For two stations i and j, define

$$U_{i,t} = F_i(X_{i,t}), \qquad U_{j,t} = F_j(X_{j,t}).$$

Then

$$U_{i,t} \sim \mathrm{Uniform}(0,1), \qquad U_{j,t} \sim \mathrm{Uniform}(0,1).$$

The marginal distributions are therefore equal after transformation, and the joint distribution of $(U_{i,t}, U_{j,t})$ contains only dependence information. $\square$

Relation between χ_{ij} and θ_{ij} under max-stability

We prove that, under bivariate max-stability with standard uniform margins, the upper tail dependence coefficient and the extremal coefficient satisfy $\chi_{ij} = 2 - \theta_{ij}$.

Proof. Let (U_i, U_j) have standard uniform margins. Assume the bivariate extreme-value copula representation

$$\Pr(U_i \le u, U_j \le u) = u^{\theta_{ij}}, \qquad u \in (0,1),$$

where

$$\theta_{ij} \in [1,2].$$

The upper tail dependence coefficient is

$$\chi_{ij} = \lim_{u \uparrow 1} \Pr(U_j > u \mid U_i > u).$$

By the definition of conditional probability,

$$\Pr(U_j > u \mid U_i > u) = \frac{\Pr(U_i > u, U_j > u)}{\Pr(U_i > u)}.$$

Since $U_i \sim \mathrm{Uniform}(0,1)$,

$$\Pr(U_i > u) = 1 - u.$$

By inclusion-exclusion,

$$\begin{aligned}
\Pr(U_i > u, U_j > u) &= 1 - \Pr(U_i \le u) - \Pr(U_j \le u) + \Pr(U_i \le u, U_j \le u) \\
&= 1 - u - u + u^{\theta_{ij}} \\
&= 1 - 2u + u^{\theta_{ij}}.
\end{aligned}$$

Therefore

$$\Pr(U_j > u \mid U_i > u) = \frac{1 - 2u + u^{\theta_{ij}}}{1 - u}.$$

Thus

$$\chi_{ij} = \lim_{u \uparrow 1} \frac{1 - 2u + u^{\theta_{ij}}}{1 - u}.$$

Both numerator and denominator converge to zero:

$$\lim_{u \uparrow 1} (1 - 2u + u^{\theta_{ij}}) = 1 - 2 + 1 = 0,$$

$$\lim_{u \uparrow 1} (1 - u) = 0.$$

By l'Hopital's rule,

$$\begin{aligned}
\chi_{ij} &= \lim_{u \uparrow 1} \frac{-2 + \theta_{ij} u^{\theta_{ij} - 1}}{-1} \\
&= \frac{-2 + \theta_{ij}}{-1} \\
&= 2 - \theta_{ij}.
\end{aligned}$$

Hence

$$\boxed{\chi_{ij} = 2 - \theta_{ij}.}$$

The endpoints are consistent with the interpretation:

$$\begin{aligned}
\theta_{ij} = 1 &\implies \chi_{ij} = 1, \\
\theta_{ij} = 2 &\implies \chi_{ij} = 0.
\end{aligned}$$

□

Extremal coefficient as the exponent measure evaluated at $(1, 1)$

We prove that, for a bivariate max-stable distribution with unit Fréchet margins, the pairwise extremal coefficient is $\theta_{ij} = V_{ij}(1, 1)$.

Proof. Let (Z_i, Z_j) denote the bivariate vector

$$(Z_i, Z_j) = (Z(s_i), Z(s_j)).$$

Assume unit Fréchet margins:

$$\begin{aligned}
\Pr(Z_i \le z) &= \exp(-1/z), \qquad z > 0, \\
\Pr(Z_j \le z) &= \exp(-1/z), \qquad z > 0.
\end{aligned}$$

Let the joint distribution have exponent measure representation

$$\Pr(Z_i \le z_i, Z_j \le z_j) = \exp\{-V_{ij}(z_i, z_j)\}, \qquad z_i, z_j > 0.$$

Set

$$z_i = z, \qquad z_j = z.$$

Then

$$\Pr(Z_i \le z, Z_j \le z) = \exp\{-V_{ij}(z, z)\}.$$

For a max-stable distribution with unit Fréchet margins, the exponent measure is homogeneous of order -1:

$$V_{ij}(az_i, az_j) = a^{-1} V_{ij}(z_i, z_j), \qquad a > 0.$$

Take

$$a = z, \qquad z_i = 1, \qquad z_j = 1.$$

Then

$$\begin{aligned}
V_{ij}(z, z) &= V_{ij}(z \cdot 1, z \cdot 1) \\
&= z^{-1} V_{ij}(1, 1).
\end{aligned}$$

Therefore

$$\begin{aligned}
\Pr(Z_i \le z, Z_j \le z) &= \exp\{-z^{-1} V_{ij}(1, 1)\} \\
&= \exp\left\{-\frac{V_{ij}(1, 1)}{z}\right\}.
\end{aligned}$$

The pairwise extremal coefficient θ_{ij} is defined by

$$\Pr(Z_i \le z, Z_j \le z) = \Pr(Z_i \le z)^{\theta_{ij}}.$$

Since
$$\Pr(Z_i \le z) = \exp(-1/z),$$
we have
$$\begin{aligned}\Pr(Z_i \le z)^{\theta_{ij}} &= [\exp(-1/z)]^{\theta_{ij}}\\ &= \exp(-\theta_{ij}/z).\end{aligned}$$
Equating the two expressions for the same joint probability gives
$$\exp\left\{-\frac{V_{ij}(1,1)}{z}\right\} = \exp\left\{-\frac{\theta_{ij}}{z}\right\}.$$
Taking logarithms gives
$$-\frac{V_{ij}(1,1)}{z} = -\frac{\theta_{ij}}{z}.$$
Multiplying by $-z$ gives
$$V_{ij}(1,1) = \theta_{ij}.$$
Hence
$$\boxed{\theta_{ij} = V_{ij}(1,1).}$$
The bounds follow from the two limiting cases. Perfect dependence gives
$$Z_i = Z_j,$$
so
$$\Pr(Z_i \le z, Z_j \le z) = \Pr(Z_i \le z) = \exp(-1/z),$$
which implies
$$\theta_{ij} = 1.$$
Independence gives
$$\begin{aligned}\Pr(Z_i \le z, Z_j \le z) &= \Pr(Z_i \le z)\Pr(Z_j \le z)\\ &= \exp(-1/z)\exp(-1/z)\\ &= \exp(-2/z),\end{aligned}$$
which implies
$$\theta_{ij} = 2.$$
Therefore
$$1 \le \theta_{ij} \le 2.$$
□

F-madogram relation to the extremal coefficient

We prove that, under bivariate max-stability with standard uniform margins, the F-madogram satisfies $\theta_{ij} = (1+2\nu_F)/(1-2\nu_F)$.

Proof. Let
$$U_i = F_i(Z_i), \qquad U_j = F_j(Z_j),$$
where U_i and U_j have standard uniform margins. The F-madogram is
$$\nu_F(i,j) = \frac{1}{2}\mathbb{E}|U_i - U_j|.$$
Use the identity
$$|a-b| = a + b - 2\min(a,b), \qquad a, b \in [0,1].$$
Then
$$\begin{aligned}\mathbb{E}|U_i - U_j| &= \mathbb{E}(U_i + U_j - 2\min(U_i,U_j))\\ &= \mathbb{E}U_i + \mathbb{E}U_j - 2\mathbb{E}\min(U_i,U_j).\end{aligned}$$
Since $U_i, U_j \sim \text{Uniform}(0,1)$,
$$\mathbb{E}U_i = \frac{1}{2}, \qquad \mathbb{E}U_j = \frac{1}{2}.$$
Therefore
$$\mathbb{E}|U_i - U_j| = 1 - 2\mathbb{E}\min(U_i,U_j).$$
Thus
$$\nu_F(i,j) = \frac{1}{2} - \mathbb{E}\min(U_i,U_j).$$

It remains to compute $\mathbb{E}\min(U_i, U_j)$. For any random variable $Y \in [0,1]$,

$$\mathbb{E}Y = \int_0^1 \Pr(Y > u)\, du.$$

Set

$$Y = \min(U_i, U_j).$$

Then

$$\begin{aligned}\mathbb{E}\min(U_i, U_j) &= \int_0^1 \Pr(\min(U_i, U_j) > u)\, du \\ &= \int_0^1 \Pr(U_i > u, U_j > u)\, du.\end{aligned}$$

By inclusion-exclusion,

$$\begin{aligned}\Pr(U_i > u, U_j > u) &= 1 - \Pr(U_i \le u) - \Pr(U_j \le u) + \Pr(U_i \le u, U_j \le u) \\ &= 1 - u - u + \Pr(U_i \le u, U_j \le u) \\ &= 1 - 2u + \Pr(U_i \le u, U_j \le u).\end{aligned}$$

Under bivariate max-stability with extremal coefficient θ_{ij},

$$\Pr(U_i \le u, U_j \le u) = u^{\theta_{ij}}.$$

Therefore

$$\begin{aligned}\mathbb{E}\min(U_i, U_j) &= \int_0^1 \{1 - 2u + u^{\theta_{ij}}\}\, du \\ &= \int_0^1 1\, du - 2\int_0^1 u\, du + \int_0^1 u^{\theta_{ij}}\, du \\ &= 1 - 2\left[\frac{u^2}{2}\right]_0^1 + \left[\frac{u^{\theta_{ij}+1}}{\theta_{ij}+1}\right]_0^1 \\ &= 1 - 1 + \frac{1}{\theta_{ij}+1} \\ &= \frac{1}{\theta_{ij}+1}.\end{aligned}$$

Hence

$$\begin{aligned}\nu_F(i,j) &= \frac{1}{2} - \frac{1}{\theta_{ij}+1} \\ &= \frac{\theta_{ij}+1}{2(\theta_{ij}+1)} - \frac{2}{2(\theta_{ij}+1)} \\ &= \frac{\theta_{ij}-1}{2(\theta_{ij}+1)}.\end{aligned}$$

Thus

$$2\nu_F(i,j) = \frac{\theta_{ij}-1}{\theta_{ij}+1}.$$

Solve for θ_{ij}:

$$\begin{aligned}2\nu_F(\theta_{ij}+1) &= \theta_{ij} - 1, \\ 2\nu_F\theta_{ij} + 2\nu_F &= \theta_{ij} - 1, \\ \theta_{ij} - 2\nu_F\theta_{ij} &= 1 + 2\nu_F, \\ \theta_{ij}(1 - 2\nu_F) &= 1 + 2\nu_F, \\ \theta_{ij} &= \frac{1+2\nu_F}{1-2\nu_F}.\end{aligned}$$

Hence

$$\boxed{\theta_{ij} = \frac{1+2\nu_F(i,j)}{1-2\nu_F(i,j)}.}$$

Replacing $\nu_F(i,j)$ by the empirical F-madogram

$$\widehat{\nu}_F(i,j;c) = \frac{1}{2B^{(c)}} \sum_{b=1}^{B^{(c)}} \left| \widehat{F}_i^{(c)}(M_{i,b}^{(c)}) - \widehat{F}_j^{(c)}(M_{j,b}^{(c)}) \right|$$

gives the estimator

$$\widehat{\theta}_{ij}^{(c)} = \frac{1 + 2\widehat{\nu}_F(i, j; c)}{1 - 2\widehat{\nu}_F(i, j; c)}.$$

□

Consistency of the empirical finite-level tail-dependence estimator

We prove that, under within-season stationarity and ergodicity, the empirical estimator $\widehat{\chi}_{ij}^{(c)}(u)$ converges to the finite-level conditional tail probability $\Pr(U_j^{(c)} > u \mid U_i^{(c)} > u)$.

Proof. Fix a season

$$c \in \{W, S\}.$$

Fix a pair of stations

$$(i, j).$$

Let

$$\{(U_{i,t}^{(c)}, U_{j,t}^{(c)}) : t \in T_c\}$$

be the season-c transformed pair process. Assume this process is strictly stationary and ergodic. Fix

$$u \in (0, 1).$$

Define

$$A_t^{(c)}(u) = \mathbf{1}\{U_{i,t}^{(c)} > u,\ U_{j,t}^{(c)} > u\},$$

and

$$B_t^{(c)}(u) = \mathbf{1}\{U_{i,t}^{(c)} > u\}.$$

Since $(U_{i,t}^{(c)}, U_{j,t}^{(c)})$ is strictly stationary and ergodic, any measurable transformation of it is also strictly stationary and ergodic. Therefore

$$\{A_t^{(c)}(u) : t \in T_c\}$$

is strictly stationary and ergodic, and

$$\{B_t^{(c)}(u) : t \in T_c\}$$

is strictly stationary and ergodic. Moreover,

$$0 \le A_t^{(c)}(u) \le 1, \qquad 0 \le B_t^{(c)}(u) \le 1.$$

Hence

$$\mathbb{E}|A_t^{(c)}(u)| < \infty, \qquad \mathbb{E}|B_t^{(c)}(u)| < \infty.$$

By the ergodic theorem,

$$\frac{1}{|T_c|} \sum_{t \in T_c} A_t^{(c)}(u) \xrightarrow{p} \mathbb{E}A_t^{(c)}(u),$$

and

$$\frac{1}{|T_c|} \sum_{t \in T_c} B_t^{(c)}(u) \xrightarrow{p} \mathbb{E}B_t^{(c)}(u).$$

By the definition of $A_t^{(c)}(u)$,

$$\begin{aligned} \mathbb{E}A_t^{(c)}(u) &= \mathbb{E}\mathbf{1}\{U_{i,t}^{(c)} > u,\ U_{j,t}^{(c)} > u\} \\ &= \Pr(U_{i,t}^{(c)} > u,\ U_{j,t}^{(c)} > u). \end{aligned}$$

By the definition of $B_t^{(c)}(u)$,

$$\begin{aligned} \mathbb{E}B_t^{(c)}(u) &= \mathbb{E}\mathbf{1}\{U_{i,t}^{(c)} > u\} \\ &= \Pr(U_{i,t}^{(c)} > u). \end{aligned}$$

Since $U_{i,t}^{(c)}$ has a standard uniform margin,

$$\Pr(U_{i,t}^{(c)} > u) = 1 - u.$$

For $u \in (0, 1)$,

$$1 - u > 0.$$

The empirical finite-level tail-dependence estimator is

$$\widehat{\chi}_{ij}^{(c)}(u) = \frac{\sum_{t \in T_c} \mathbf{1}\{U_{i,t}^{(c)} > u,\ U_{j,t}^{(c)} > u\}}{\sum_{t \in T_c} \mathbf{1}\{U_{i,t}^{(c)} > u\}}.$$

Using $A_t^{(c)}(u)$ and $B_t^{(c)}(u)$,

$$\begin{aligned}\widehat{\chi}_{ij}^{(c)}(u) &= \frac{\sum_{t \in T_c} A_t^{(c)}(u)}{\sum_{t \in T_c} B_t^{(c)}(u)} \\ &= \frac{|T_c|^{-1} \sum_{t \in T_c} A_t^{(c)}(u)}{|T_c|^{-1} \sum_{t \in T_c} B_t^{(c)}(u)}.\end{aligned}$$

By the continuous mapping theorem,

$$\begin{aligned}\widehat{\chi}_{ij}^{(c)}(u) &\xrightarrow{p} \frac{\Pr(U_{i,t}^{(c)} > u,\ U_{j,t}^{(c)} > u)}{\Pr(U_{i,t}^{(c)} > u)} \\ &= \Pr(U_{j,t}^{(c)} > u \mid U_{i,t}^{(c)} > u).\end{aligned}$$

Therefore

$$\boxed{\widehat{\chi}_{ij}^{(c)}(u) \xrightarrow{p} \Pr(U_{j,t}^{(c)} > u \mid U_{i,t}^{(c)} > u)}$$

for every fixed $u \in (0, 1)$. If the limit

$$\chi_{ij}^{(c)} = \lim_{u \uparrow 1} \Pr(U_{j,t}^{(c)} > u \mid U_{i,t}^{(c)} > u)$$

exists, then the finite-level probability estimated above is the finite-u approximation to the asymptotic tail-dependence coefficient. $\square$

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Table 1: All available variables in `hourly-in-situ-meteorological-observations`.

Parameter	Description	Unit	Observed property	Comment
DD	Past 10 minute mean wind direction, representative for 10 metres, in degrees; 360 = north; 90 = east; 180 = south; 270 = west; 0 = calm; 990 = variable	°	Wind from direction	
DR	Hourly precipitation duration, in hours	h	Duration of precipitation	
EE	Hourly mean vapour pressure, in hectopascal	hPa	Water vapour partial pressure in air	
FF	Past 10 minute mean wind speed, representative for 10 metres, in metres per second	m/s	Wind speed	
FH	Hourly mean wind speed, representative for 10 metres, in metres per second	m/s	Wind speed	
FX	Hourly maximum wind gust, representative for 10 metres, in metres per second	m/s	Wind speed of gust	
IX	Hourly indicator present weather code; 1 = manned and recorded using code from visual observations; 2; 3 = manned and omitted, no significant weather phenomenon to report or not available; 4 = automatically recorded using code from visual observations; 5; 6 = automatically omitted, no significant weather phenomenon to report or not available; 7 = automatically set using code from automated observations	–	Indicator present weather code	
N	Past 10 minute cloud cover, in okta; 9 = sky invisible	okta	Cloud area fraction	
P	Past 1 minute mean air pressure at sea level	hPa	Air pressure at mean sea level	
Q	Hourly global solar radiation, in joules per square centimetre	J/cm^2	Integral with respect to time of surface downwelling shortwave flux in air	
RH	Hourly precipitation amount, in millimetres	mm	Precipitation amount	[1]
SQ	Hourly sunshine duration, calculated from global solar radiation, in hours	h	Duration of sunshine	[2]
T	Past 1 minute air temperature at 1.50 metres, in degrees Celsius	°C	Air temperature	
T10N	Past 6 hour minimum air temperature at 10 centimetres, in degrees Celsius	°C	Air temperature	
TD	Past 1 minute dew point temperature at 1.50 metres, in degrees Celsius	°C	Dew point temperature	
U	Past 1 minute relative atmospheric humidity at the time of observation, at 1.50 metres, as percentage	%	Relative humidity	
VV	Past 10 minute horizontal visibility; 0 = less than 100 m; 1 = 100–200 m; 2 = 200–300 m; . . . ; 49 = 4900–5000 m; 50 = 5–6 km; 56 = 6–7 km; 57 = 7–8 km; . . . ; 79 = 29–30 km; 80 = 30–35 km; 81 = 35–40 km; . . . ; 89 = more than 70 km	–	Visibility in air	
W1	Hourly indicator of fog; 0 = no occurrence; 1 = occurred	–	Fog	
W2	Hourly indicator of rainfall; 0 = no occurrence; 1 = occurred	–	Rain	
W3	Hourly indicator of snow; 0 = no occurrence; 1 = occurred	–	Snow	
W5	Hourly indicator of thunder; 0 = no occurrence; 1 = occurred	–	Thunder	
W6	Hourly indicator of ice formation; 0 = no occurrence; 1 = occurred	–	Ice formation	
WW	Hourly present weather code, aggregated with an algorithm using 10-minute `ww` WaWa values from the preceding 6 hours; WMO table 4680	–	Present weather code	

Notes: [1] For precipitation amounts the detection minimum is 0.05 millimetres. [2] For sunshine duration the measured minimum is 0.05 hours.